

OPERATIONS PLANNING AND CONTROL-ME354

MINI PROJECT:

Improving Patient Scheduling During COVID-19 Surges Using Hybrid Priority Rules

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Abstract

During the COVID-19 second wave in India, hospitals faced a severe shortage of resources such as ICU beds and oxygen support, making patient scheduling a critical challenge. This study models the hospital system as a resource-constrained scheduling problem, where each patient is treated as a task with a processing time and a due date representing urgency. A dataset of 50 patients was generated based on severity levels (Critical, Severe, Moderate, Mild), and four scheduling methods—FCFS, SPT, EDD, and a Hybrid rule ($\alpha = 0.4$)—were analyzed. Performance was evaluated using mean flow time, mean tardiness, maximum tardiness, and a combined performance index. The results show that SPT provides the lowest mean flow time and tardiness, indicating high system efficiency; however, it fails to reduce extreme delays, which is critical in healthcare. EDD performs poorly due to ignoring processing time, leading to higher delays. The hybrid approach gives balanced results by considering both urgency and processing time, reducing extreme delays while maintaining reasonable performance. This study highlights the trade-off between efficiency and fairness in patient scheduling and recommends hybrid scheduling for real-world healthcare systems, as it better ensures timely treatment of critical patients while maintaining efficient resource utilization.

Introduction

During the second wave of COVID-19 in India (April–May 2021), hospitals faced an unprecedented surge in critically ill patients. Tertiary care centers such as Gandhi Hospital were admitting more than 80 severe cases daily, while operating with limited resources, including around 300 ICU beds and 200 oxygen-supported beds. This imbalance between demand and capacity created a situation where not every patient could receive immediate care, forcing hospitals to decide the order in which patients should be treated.

To analyze this challenge, the hospital system can be viewed as a resource-constrained setup where patients arrive continuously and compete for limited treatment facilities. In this context, each patient is treated as a task that must be handled within a critical time window. Unlike industrial systems, where delays may lead to financial losses, delays in healthcare directly impact patient survival. Studies indicate that even a one-hour delay in receiving appropriate care can significantly increase mortality risk.

Traditional scheduling approaches, such as treating patients in arrival order or prioritizing shorter treatments, fail to capture the urgency and complexity of medical conditions. These methods may unintentionally delay critical patients. Therefore, this study focuses on modeling patient flow and resource allocation more realistically to minimize delays and improve survival outcomes.

Research Gap and Objectives

Research Gap

Existing research on hospital scheduling mainly focuses on efficiency, such as reducing waiting time or improving resource use, but does not fully consider life-and-death urgency during crises like COVID-19. Most models use fixed data or handle problems separately, while real hospitals face unpredictable patient arrivals and limited resources. Common methods like FCFS and fixed priority rules fail under heavy load, causing delays and increased mortality. Even advanced models are often too complex for real-time use. Hence, there is a clear gap for a simple, practical method that balances patient urgency and system efficiency together in real-time emergency situations.

Research Objectives

The main objective of this project is to improve how hospitals allocate limited resources like ICU and oxygen beds during high patient inflow situations such as COVID-19. It aims to compare different scheduling methods including FCFS, SPT, EDD, and a proposed Hybrid rule to understand their impact on patient waiting time and survival. The study focuses on developing a balanced approach that considers both patient urgency and treatment time. Another objective is to find the optimal weight value for the Hybrid method to minimize delays and reduce mortality risk, ultimately helping hospitals make better real-time decisions under extreme conditions.

Data Collection and Generation

The system is based on the situation at Gandhi Hospital, Hyderabad, during the COVID-19 second wave, where a large number of patients were coming and resources were limited. Because of this, proper scheduling became very important. Patients are divided into four categories—Critical, Severe, Moderate, and Mild—based on their oxygen levels (SpO_2). For each patient, two values are considered: processing time (p), which means how many days treatment is needed, and due date (d), which means how quickly the patient should receive treatment.

Since real patient data is not available due to privacy reasons, we created a dataset of 50 patients. For each patient, we first assigned a severity category and then selected values of p and d within realistic ranges for that category. This method helps to keep the data close to real-life conditions while also adding some variation. The dataset shows the balance between urgency and treatment time, which is important for comparing different scheduling methods.

Table 1: Severity-wise Clinical Parameters for Patient Scheduling Model

Severity	SpO2	Resource	p (days)	d (survival window)
Critical	< 85%	ICU/Ventilator	16 - 18	4 - 12 Hours
Severe	85-90%	Oxygen Bed	11 - 13	12 - 24 Hours
Moderate	90-93%	Oxygen Bed	8 - 10	2 - 3 Days
Mild	> 93%	General Ward	5 - 7	14 Days

Methodology and Calculation:

The study evaluates the scheduling of 50 patients using four different rules: First Come First Serve (FCFS), Shortest Processing Time (SPT), Earliest Due Date (EDD), and a Hybrid rule with $\alpha = 0.4$. Each patient is characterized by a processing time and a due date, where the due date represents urgency. For each scheduling rule, patients are arranged according to the respective priority logic. In the hybrid approach, a combined priority index is calculated as $0.4p_i + 0.6d_i$, giving higher importance to urgency while still considering processing time.

Assumptions

The following assumptions are made in this study:

- All patients arrive at time $t=0$ (no dynamic arrivals)
- Only one patient is treated at a time (single-server system)
- Preemption is not allowed (once treatment starts, it must finish)
- Processing times and due dates are known in advance
- Severity is indirectly represented through due dates
- The objective is to balance efficiency and urgency

Four scheduling rules are applied:

- FCFS: Patients are processed in given order
- SPT: Patients are sorted in ascending order of processing time p_i
- EDD: Patients are sorted in ascending order of due date d_i
- Hybrid: A priority index is computed as

$$Priority_i = 0.4 p_i + 0.6 d_i$$

Patients are then sorted in ascending order of this index

For each sequence, start time, completion time, flow time, and tardiness are computed. The first patient is assumed to start at time zero, and subsequent start times are equal to the completion time of the previous patient. Flow time is taken as the completion time, and tardiness is calculated as the positive difference between completion time and due date. Based on these values, mean flow time, mean tardiness, and maximum tardiness are obtained for each scheduling rule.

For a given sequence, the following calculations are performed:

$$S_1=0, S_i=C_{i-1}$$

$$C_i=S_i+p_i$$

$$F_i=C_i$$

$$T_i=\max(0, C_i-d_i)$$

To enable comparison among different methods, a combined performance index is defined as $J=0.4(\text{Mean Flow Time})+0.6(\text{Mean Tardiness})$. The higher weight assigned to tardiness reflects the importance of urgency in healthcare scenarios, where delays can have critical consequences.

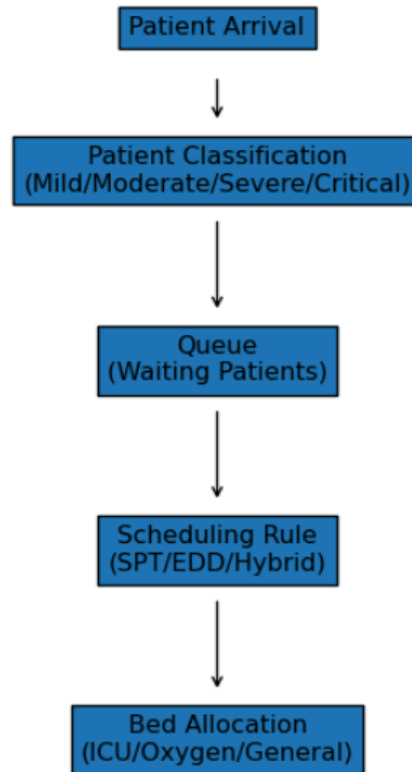


Figure 5.3: System model for patient scheduling and bed allocation

The following algorithm describes the step-by-step implementation of the proposed weighted hybrid scheduling rule used for patient prioritization and bed allocation.

Algorithm: Hybrid Priority Scheduling

Step 1: Initialize system time $t=0$ and input patient dataset (p_j, d_j)

Step 2: For each patient j , compute *slack*: $slack_j = d_j - t - p_j$

Step 3: Compute priority score: $Score_j = \alpha \cdot p_j + (1 - \alpha) \cdot slack_j$

Step 4: Sort all patients in ascending order of $Score_j$.

Step 5: Select the patient with the lowest score and allocate appropriate bed

Step 6: Update completion time: $t = t + p_j$

Step 7: Remove allocated patient from queue

Step 8: Repeat Steps 2–7 until all patients are scheduled

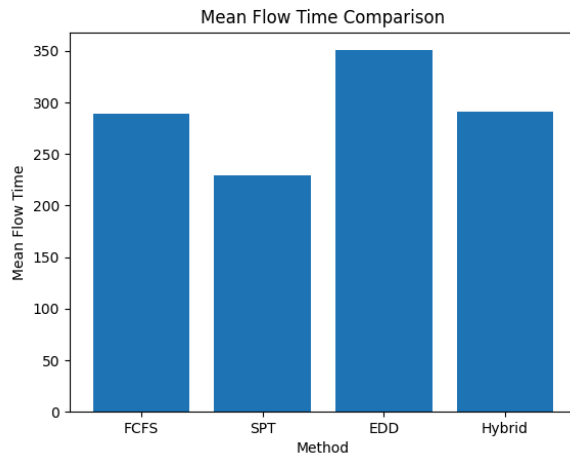
This algorithm ensures a balance between urgency (slack time) and system efficiency (processing time), enabling improved survival outcomes during high-demand scenarios.

Results and Discussion

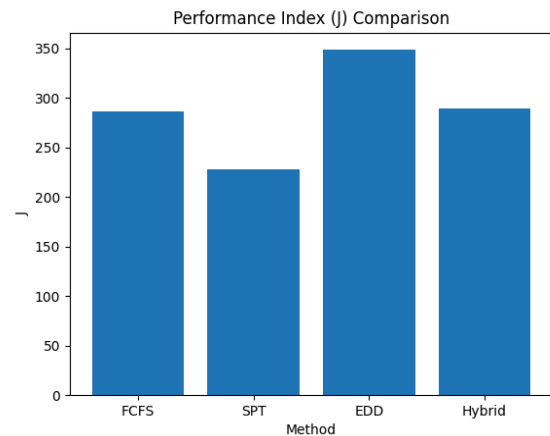
The comparative performance of the four scheduling methods is presented in Table 1. The evaluation is based on mean flow time, mean tardiness, maximum tardiness, and the composite performance index J .

Table 2: Comparative performance of the four scheduling methods

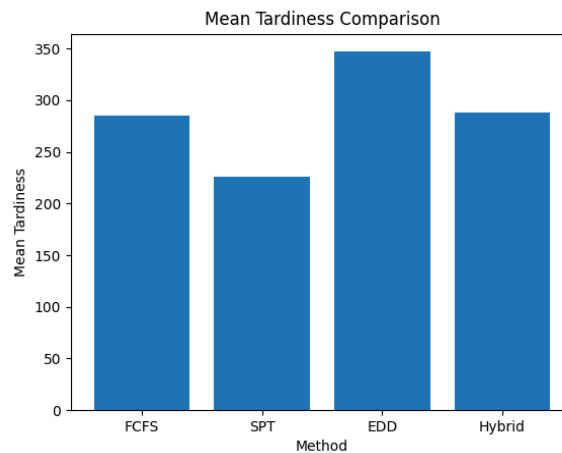
Rule	Mean Flow Time	Max Tardiness	Mean Tardiness	Performance Index
FCFS	288.73	569.45	285.01	286.49
SPT	229.66	569.46	226.16	227.56
EDD	350.75	555.4	347.03	348.52
Hybrid ($\alpha=0.4$)	291.42	555.4	287.70	289.19



Graph 1: Mean Flow Time vs Method



Graph 2: Performance Index vs Method



Graph 3: Mean Tradiness vs Method

Discussion

To better visualize the results, bar charts of Mean Flow Time, Mean Tardiness, and Performance Index (J) were plotted for all methods. These charts clearly show the comparative differences in system performance.

The results indicate a clear variation across scheduling rules. When moving from FCFS to SPT, both mean flow time and mean tardiness decrease significantly. This change occurs because SPT prioritizes shorter processing times, allowing faster completion of tasks and reducing overall waiting time. However, maximum tardiness remains high, indicating that some patients still face extreme delays.

In contrast, EDD shows a sharp increase in both mean flow time and mean tardiness. This happens because EDD schedules patients only based on urgency, often placing long-duration cases earlier, which slows down the system and increases delays for others. Although it slightly reduces maximum tardiness, its overall performance is poor.

The hybrid method shows moderate values, balancing both processing time and urgency. It avoids the extremes seen in SPT and EDD and slightly reduces maximum tardiness compared to FCFS and SPT.

From an operational perspective, these results highlight a trade-off between efficiency and fairness. SPT improves efficiency but ignores urgency, while EDD over-prioritizes urgency and reduces efficiency. The hybrid approach offers a balanced solution, making it more suitable for practical applications where both timely service and fairness are important.

Conclusion and Recommendations

This study compared four scheduling methods—FCFS, SPT, EDD, and Hybrid—based on mean flow time, mean tardiness, maximum tardiness, and a combined performance index. The results show that SPT achieves the best numerical performance by minimizing both mean flow time and mean tardiness, indicating high system efficiency. However, it does not reduce extreme delays, as reflected in the high maximum tardiness.

EDD, although designed to prioritize urgency, performs poorly in overall system efficiency due to ignoring processing time, leading to increased delays. FCFS also shows weak performance as it does not consider either urgency or processing time.

The hybrid approach provides a balanced performance by considering both processing time and urgency. While it does not achieve the lowest numerical score, it reduces extreme delays and offers more stable system behavior.

From a practical perspective, it is recommended to use hybrid scheduling in real-world applications, especially in healthcare systems, where both efficiency and timely treatment are important. Future work can improve this model by incorporating severity-base

Github link:

[Surekha2006/Surekha_Suchith_Lakshmi: Patient scheduling optimization using FCFS, SPT, EDD, and Hybrid methods under resource constraints](#)

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